C5 Energy Changes – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Exothermic and Endothermic **Paper 1**

#	Question	Answer
1	Exothermic reaction	Transfers energy to the surroundings; temperature rises
2	Endothermic reaction	Takes in energy from the surroundings; temperature falls
3	Examples of exothermic reactions	Combustion, oxidation, neutralisation
4	Examples of endothermic reactions	Thermal decomposition, citric acid + sodium hydrogencarbonate
5	Uses of exothermic reactions	Hand warmers, self-heating cans
6	Use of endothermic reactions	Sports injury cold packs
7	Activation energy	Minimum energy particles need to react
8	Bond breaking: endothermic or exothermic	Endothermic
9	Bond making: endothermic or exothermic	Exothermic
10	Energy change from bond energies	Bonds broken – bonds made

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AQA GCSE Chemistry • Set 2: Reaction Profiles and Measuring Energy Paper 1

#	Question	Answer
1	Products on an exothermic profile	Lower than reactants
2	Products on an endothermic profile	Higher than reactants
3	Sign of energy change: exothermic	Negative
4	Sign of energy change: endothermic	Positive
5	Conservation of energy	Energy is not created or destroyed
6	Container for measuring temperature changes	Polystyrene cup with lid
7	Why polystyrene is used	Good insulator
8	Why a lid is used	Reduces heat loss
9	Main source of error	Heat loss to surroundings
10	Units of bond energy	kJ/mol

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AQA GCSE Chemistry • Set 3: Chemical Cells and Fuel Cells Paper 1 Triple only

#	Question	Answer
1	Simple chemical cell	Two different metals in an electrolyte
2	Factors affecting cell voltage	Metals used and the electrolyte
3	Why non-rechargeable cells stop working	Reactants are used up
4	Rechargeable cell	Reactions reversed by an external current
5	What a hydrogen fuel cell is supplied with	Hydrogen and oxygen (air)
6	Product of a hydrogen fuel cell	Water
7	Overall equation	$2H_2 + O_2 \rightarrow 2H_2O$
8	Half-equation at the negative electrode	$2H_2 \rightarrow 4H^+ + 4e^-$
9	Half-equation at the positive electrode	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
10	Disadvantage of hydrogen fuel cells	Hydrogen is hard to store and flammable